

RESEARCH GAP IDENTIFICATION

Project Title

Machine Learning-Based Loan Risk Prediction: A Comparative Study of Ensemble, Oversampling, and Topological Approaches for Credit and Loan Default Assessment

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1. Limited Model Interpretability

Most of the reviewed studies, including Zhuang and Wei (2024), Zhao (2024), Yotsawat et al. (2021), and Sayed et al. (2024), focus primarily on maximizing predictive accuracy rather than explaining how individual borrower or loan features contribute to a default prediction. Only one of the five reviewed papers applied an explainability technique (SHAP), leaving most models to function as black boxes that offer little transparency for loan officers, regulators, or borrowers.

2. Single-Domain and Single-Dataset Validation

Each reviewed study was developed and validated within a single, narrow lending domain, such as personal consumer credit, medium/micro enterprise credit, or maritime finance, and mostly on one dataset. This limits confidence in how well the proposed models would generalize across different types of borrowers, loan products, or lending institutions.

3. Trade-offs in Imbalance-Handling Techniques Remain Unresolved

Different studies adopt different strategies for class imbalance, including GAN-based oversampling (WGAN-GP), SMOTE and SMOTE-TOMEK, and cost-sensitive learning, but no study directly compares these approaches against one another on the same dataset. As a result, it is unclear which imbalance-handling strategy is most effective and efficient for loan default prediction, and each technique carries its own limitation, such as unrealistic synthetic samples, boundary noise, or the need for manually tuned misclassification costs.

4. Linear Models Cannot Capture Complex Financial Relationships

Zhao's (2024) Linear Learning Analysis approach offers simplicity and interpretability but is explicitly limited when applied to non-linear, high-frequency financial data. This highlights a gap between interpretable-but-limited linear methods and accurate-but-opaque non-linear ensemble or deep learning methods, with few studies attempting to bridge the two.

5. High Computational Cost of Deep Learning Approaches

Sayed et al. (2024) found that deep learning architectures such as DenseNet and ResNet achieved the highest accuracy but required substantial computational resources. This makes such models less practical for smaller financial institutions or real-time loan approval systems, and existing literature does not adequately address this trade-off between accuracy and deployability.

6. Exclusion of Non-Financial and Contextual Factors

Kheneifar and Amiri (2025) explicitly noted that their maritime finance model excluded broader economic and geopolitical conditions that can influence default risk. More generally, the reviewed studies rely almost entirely on transactional and financial features, with little exploration of external contextual factors that could improve prediction robustness.

7. Domain-Specific Techniques Are Not Broadly Applicable

The Topological Data Analysis and Graph Neural Network approach used by Kheneifar and Amiri (2025) achieved strong results but was designed specifically for maritime finance and is computationally intensive. Its applicability to other lending domains, such as personal or SME credit, has not been examined in the literature.

8. Lack of a Unified Comparative Framework

No single study in the reviewed literature evaluates ensemble learning, oversampling strategies, and topological or graph-based approaches together within one comparative framework. Each paper focuses on a different combination of domain, algorithm, and imbalance-handling method, making it difficult to draw direct conclusions about which overall approach performs best for loan risk prediction.

9. Absence of Practical, Decision-Support Oriented Systems

The reviewed models remain largely at the research and evaluation stage, focused on reporting accuracy, AUC, and related metrics, rather than being developed into decision-support tools that loan officers or financial institutions could use directly, complete with transparent, feature-level justifications for each prediction.

Conclusion of Research Gap

From the identified research gaps, it is evident that there is a need for a comparative loan risk prediction framework that evaluates ensemble learning, oversampling, and topological approaches under a common setting, incorporates explainable AI techniques such as SHAP for transparent decision support, and is validated across multiple lending domains and datasets. Such a framework would improve prediction reliability, provide interpretable outputs for financial institutions, and support more informed and trustworthy loan approval decisions.